



Sharing the intel

It is not just companies that should be aware of IoT security concerns, but even the end users. <https://dgit.in/apr20-83>



Russia wanted its own Mirai

A hacking group has revealed plans, allegedly by Russia, of building their own Mirai style botnet. <https://dgit.in/apr20-84>

for the devices can be viewed here: <https://dgit.in/dlinkpoc>

The Internet of Things can be easy to exploit in cases where the default passwords aren't changed or obvious flaws are present but most of the time it is difficult because the amount of experience required for doing reverse engineering and binary exploitation is quite hefty. But hey, you need to start somewhere! With the knowledge about the methodology and appropriate toolset, you can pentest and play around with any device you can get your hands on. There are online IoT CTFs available so you can hone your skills if you don't have access to actual hardware devices. Check out these awesome resources, all of them focussed on IoT security which can prove to be a jackpot for those who are interested!

- nebgahz's list of Awesome IoT Hacks: <https://dgit.in/nebgahz>
 - V33RU's IoT Security and Pentesting 101: <https://dgit.in/v33ru>
 - flkie-cad's awesome embedded and IoT security: <https://dgit.in/flkiecad>
- All of these have a curated list of IoT security tools, resources, tutorials, white papers, conference videos and everything else that you might need!

THE MITIGATIONS:

There is no denying that IoT security is a hot mess, but you don't have to be a victim of that. First of all, vendors should acknowledge that lax security in IoT devices is indeed a huge problem and not something that can be sidelined. Classic vulnerabilities in core development or functionality implementation can be done away with secure coding practices, which always has proved fruitful. Building devices on top of older and legacy software systems needs to be halted, and replaced with modern operating systems and software. Even then, regular firmware updates and security patches are crucial. Today most devices need manual updation, which we think needs to be changed where crucial updates are applied automatically anytime the device is connected to the internet, without

user intervention. This will make sure the device is secure even if the user was unaware of the existence of a vulnerability or the patch to that vulnerability. Vendors need to ditch the mentality that IoT devices are "unworthy" of hacks and consider their security a liability, for they can render whole infrastructures vulnerable.

But vendors are not always to be trusted. As a user, you can't control the preventive measures being taken by the creators and companies that market the devices. That doesn't mean you are helpless against various breach of privacy and attacks. There are many changes you can implement in your usage of these devices to counter the adversaries:

- As soon as you receive any device connected to the internet, change the default passwords. That doesn't mean you change "admin:admin"

tive data like banking credentials away from the reach of these devices. Keep them in a secure and trusted device or on your system which is protected.

- Use VPN to encrypt your data that is being transmitted over your home network or a public Wi-Fi. If a VPN was installed on the router, the TRENDnet hack could've been prevented.
- Keeping personal devices away from workplaces is a good way to not compromise the security of the organisation you work for. Otherwise, it can cause a huge loss for the company and can create a lot of problems for you.

If you close your eyes for a minute and look at your life five years from now, you'll find yourself connected to more devices than ever. Your ceiling fans will regulate themselves accord-

```
eve@eve:~$ ./exploit.py 'ls -l /'
drwxr-xr-x  2 1000  1000  4096 Jan 14 14:16 bin
drwxrwxr-x  3 1000  1000  4096 May  9 16:04 dev
drwxrwxr-x  3 1000  1000  4096 Sep  3 20:10 etc
drwxrwxr-x  3 1000  1000  4096 Jan 14 14:16 lib
drwxr-xr-x  3 1000  1000  4096 Jan 14 14:16 libexec
lrwxrwxrwx  1 1000  1000    11 May  9 16:01 linuxrc ->
drwxrwxr-x  2 1000  1000  4096 Nov 11 2008 lost+found
drwxrwxr-x  7 1000  1000  4096 May  9 15:44 mnt
drwxr-xr-x  2 1000  1000  4096 Jan 14 14:16 mydlink
drwxrwxr-x  2 1000  1000  4096 Nov 11 2008 proc
drwxrwxr-x  2 1000  1000  4096 May  9 17:49 root
drwxr-xr-x  2 1000  1000  4096 Jan 14 14:16/sbin
drwxrwxr-x  3 1000  1000  4096 May 15 04:27 tmp
drwxrwxr-x  7 1000  1000  4096 Jan 14 14:16 usr
drwxrwxr-x  3 1000  1000  4096 May  9 16:04 var
-rw-r--r--  1 1000  1000    17 Jan 14 14:16 version
drwxrwxr-x  8 1000  1000  4096 May  9 16:52 www
```

Fig. Exploring the device using shell commands

to "user:123456789". Always ensure your passwords are something that are not brute-forceable and cannot be found in a dictionary of passwords.

- Do your homework before buying a smart device. They are smart because they collect a huge amount of data. But what's happening with data is not necessarily always something you would want. Research the product's privacy policies and terms on how the data is stored, protected or if it is being shared with any third parties.
- Be wary and keep highly sensi-

ing to the temperature, refrigerators will be popping notifications about food being rotten and you won't need a key to unlock the doors anymore. But all this is possible if, and it's a big if, the security is strengthened to such an extent that IoT devices become a norm and we can trust them as we trust our padlock. Our whole lives will soon depend on technology and the drift is already noticeable with smart healthcare devices, automobiles, etc. If not, a crisis awaits us because as we get more digital so will the thieves and it's harder to stop the digital ones. So the question looms, to IoT or not to? 